Physical Chemistry CUME Sept. 6th, 2019

Thermodynamic Laws and Entropy

1. State the First Law of Thermodynamics

2. State the Second Law of Thermodynamics

3. State the Third Law of Thermodynamics

4. The Entropy change is described by the Clausius Inequality, $dS \geq \frac{\delta Q}{T}$, please specify whether each variable is a state function or a path function, whether each variable is an intensive property or extensive property. When will the equal sign hold in the Clausius Equation?

5. What is (roughly) the heat capacity of liquid water? Compute the entropy change for water if it is cooled by 1 K from ambient temperature of 25 C.

6. Draw schematically the entropy *S* as a function of temperature for water from 263 K to 293 K.